



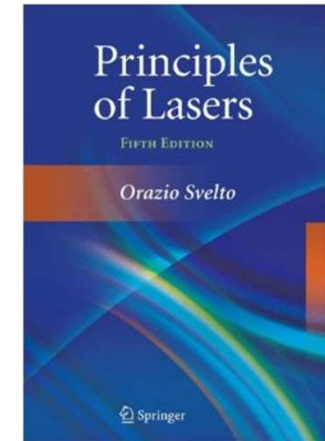
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch3-1

Resonant Interaction between Electromagnetic Field & Matter



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Chapter 3: Resonant Interaction between Electromagnetic Field & Matter

❖ 3.1 Light and matter interaction

❖ 3.2 Line broadening mechanisms

- Homogeneous broadening: natural, collisions, lattice vibration
- Inhomogeneous broadening: Doppler, lattice defect
- Comprehensive broadening

3.1 Light & Matter Interaction

The physical basis of laser: The resonant interaction between electromagnetic fields and the (bound) electron inside atoms (or ions , molecules) which form the matter.

Laser theory: Light & Matter

1) The Classical Theory – Classical atomic luminous model

- Description of light: classical electromagnetic field (Maxwell equations)
- Description of atom: classical atomic model (dipole resonator)

You can approximate describe absorption, dispersion, spontaneous emission and spontaneous emission line-width and other physical phenomena, can not describe the nonlinear physical process (saturated, nonlinear polarization, etc.)

2) **Semi-Classical Theory** – the Lamb theory (*W. E. Lamb Jr*, 1964)

- Description of light: classical electromagnetic field (Maxwell equations)
- Description of atom: quantum mechanics

Can deal with the volatility of light and the relevant physical phenomena like non-linear phenomena and Lamb dip, etc., but can not deal with the related issues of the particle nature of light (quantum optics), such as the quantum fluctuations of light, and the photon statistics.

3) **Quantum Theory** – quantum electrodynamics (QED) using quantum processing to deal with radiation field and atom

- Description of light: photons by using QED
- Description of atom: QED model

It is the foundation of modern quantum optics, the physical problems associated with particles of light can be handled, but very complicate in handling issues related to the volatility of light (such as phase). In QED, the number of photons (i.e. light amplitude) and the phase is a quantity of uncertainty.

4) Rate Equation – a simplified quantum theory

Interaction between electromagnetic field (photons) & atoms of gain medium

Does not consider the quantum fluctuations of the number of photons and phase of the light, only discuss the number of photons (light intensity).

The starting point of rate equation: the basic relationship of SP, STA and STE

$$\left. \begin{aligned} A_{21} &= \left(\frac{dn_{21}}{dt} \right)_{sp} \frac{1}{n_2} \\ W_{12} &= \left(\frac{dn_{12}}{dt} \right)_{sta} \frac{1}{n_1} \\ W_{21} &= \left(\frac{dn_{21}}{dt} \right)_{ste} \frac{1}{n_2} \end{aligned} \right\} \longrightarrow \left\{ \begin{aligned} \left(\frac{dn_{21}}{dt} \right)_{sp} &= A_{21} n_2 \\ \left(\frac{dn_{12}}{dt} \right)_{sta} &= W_{12} n_1 \\ \left(\frac{dn_{21}}{dt} \right)_{ste} &= W_{21} n_2 \end{aligned} \right.$$

$$\begin{aligned} W_{12} &= B_{12} \rho_\nu \\ W_{21} &= B_{21} \rho_\nu \\ B_{12} f_1 &= B_{21} f_2 \\ \rho_\nu &= \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\frac{h\nu}{k_b T}} - 1}
 \end{aligned}$$

single frequency $\nu = (E_2 - E_1)/h \longrightarrow P(\nu)$ has a distribution

3.2 Line broadening mechanisms

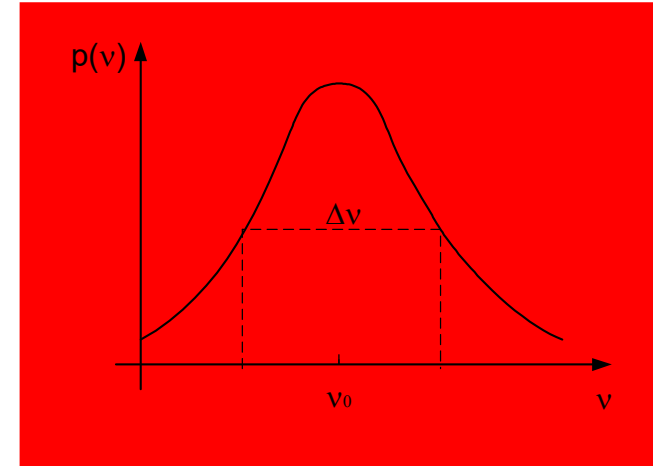
1. Line broadening $P(\nu)$

$$P = \frac{dn_{21}}{dt} h\nu = n_2 A_{21} h\nu$$

$$P = \int_{-\infty}^{+\infty} P(\nu) d\nu$$

2. Line function: spectrum line-shape $\tilde{g}(\nu, \nu_0)$

$$\tilde{g}(\nu, \nu_0) = \frac{P(\nu)}{P} \quad \int_{-\infty}^{+\infty} \tilde{g}(\nu, \nu_0) d\nu = 1$$



3. Spectrum line-width $\Delta\nu$

4. Two broadening mechanisms: homogenous and inhomogeneous broadening

I. Homogenous Broadening

natural, collisions, lattice vibration (phonon)

1) **Natural broadening:** atoms have limited lifetime in the excited state

The dipole oscillator damping vibration

$$x(t) = x_0 e^{-\frac{\gamma t}{2}} e^{i\omega_0 t} \xrightarrow{\text{FT}} x(\nu) = \frac{x_0}{\frac{\gamma}{2} - i(\nu_0 - \nu)2\pi}$$

$$P(\nu) d\nu \propto |x(\nu)|^2 d\nu$$

$$P = \int_{-\infty}^{+\infty} P(\nu) d\nu$$

Normalization condition

$$\tilde{g}_N(\nu, \nu_0) = \frac{P(\nu)}{P} = \frac{\gamma}{(\gamma/2)^2 + 4\pi^2(\nu - \nu_0)^2}$$

The damping coefficient γ and spontaneous emission lifetime τ_s

(1) According to the concept of damping vibration

$$P(t) \propto n_{20} |x(t)|^2 = n_{20} x_0^2 e^{-\gamma t} = P_0 e^{-\gamma t}$$

(2) According to the change of particles number in the E_2 state

$$\frac{dn_2}{dt} = -\frac{dn_{21}}{dt} = -\frac{n_2}{\tau} \Rightarrow n_2(t) = n_{20} e^{-t/\tau}$$

$$n_2(t) = n_{20} e^{-t/\tau} \quad P(t) = P_0 e^{-\gamma t}$$

$$P(t) = -\frac{dn_2(t)}{dt} h\nu = \underbrace{n_{20} h\nu A_{21}}_{P_0} e^{-t/\tau_s} = P_0 e^{-t/\tau_s}$$

$\gamma = 1/\tau_s$

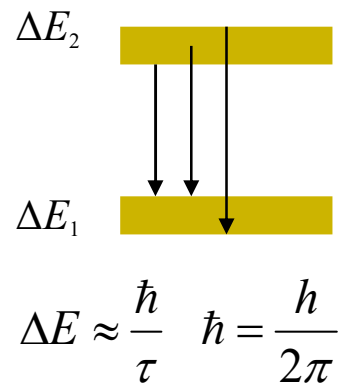
Lorentz function

$$\tilde{g}_N(\nu, \nu_0) = \frac{1/\tau_s}{(1/2\tau_s)^2 + 4\pi^2(\nu - \nu_0)^2}$$

$$\tilde{g}_N(\nu, \nu_0) = \frac{\Delta\nu_N/2\pi}{(\nu - \nu_0)^2 + (\Delta\nu_N/2)^2}$$

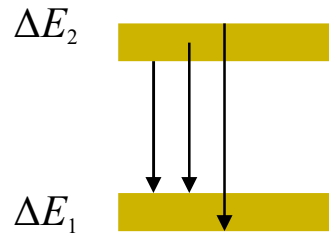
$$\Delta\nu_N = \frac{1}{2\pi\tau_s} \quad \text{natural broadening line-width}$$

The quantum interpretation of line broadening due to the limited lifetime of energy level

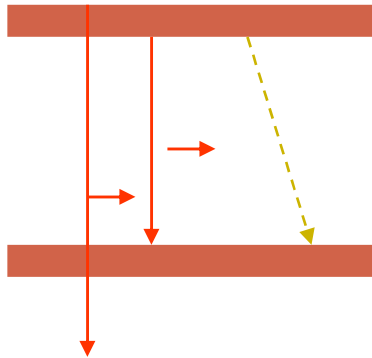


Fluorescence - energy level transition between quantum states

- The average lifetime of energy level can be understood as atoms in the state has an uncertainty of time τ (time is not a determined value);
- Time and energy can't be accurately measured at the same time, the state corresponding to the energy also have a uncertainty of ΔE .



If the high and low energy levels have certain width, the transition between them is not defined by a certain frequency ν_{21} , but the one with a range of frequency distribution $\Delta\nu$



$$\Delta\nu_N = \frac{\Delta E_2 + \Delta E_1}{h} \approx \frac{1}{2\pi\tau_1} + \frac{1}{2\pi\tau_2}$$

$$\frac{1}{\tau} = \frac{1}{\tau_{sp}} + \frac{1}{\tau_{nr}}$$

Ne atom, 632.8 nm, energy life time is 10^{-8} s, natural broadening 10^7 Hz

CO₂ molecule, 10.6 μm , natural broadening 10^3 - 10^4 Hz

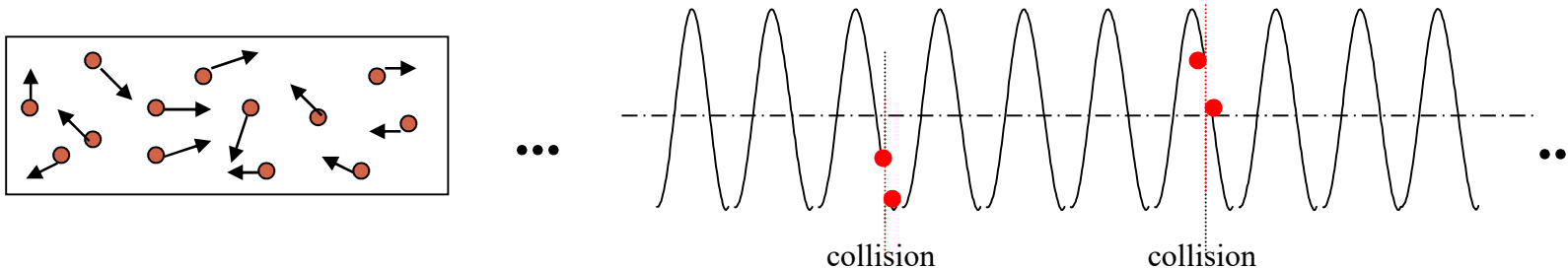
$$\Delta\nu_N = \frac{1}{2\pi\tau_s}$$

Completely ignore the width of the grand state energy level, this is due to the limitation of the classical radiation model.

2) Collision Broadening: random "collision" between atoms

- Inelastic collision: internal energy transfer, equivalent to excited state lifetime ↓
ground state atom ↔ excited state atom
excited state atom ↔ other atoms
- Elastic collision: spontaneous radiation & phase change, length ↓

Due to the random collisions, the phase change is big, and it is not related to the interrupted wave. The average length of the wave is determined by the average time



$$\Delta \nu_L = \frac{1}{\pi \tau_L}$$

$\Delta \nu_L$ — collision line-width

τ_L — average collision time (the time spacing between collision)

$$\tilde{g}_L(\nu, \nu_0) = \frac{\Delta\nu_L/2\pi}{(\nu - \nu_0)^2 + (\Delta\nu_L/2)^2} \quad \Delta\nu_L = \frac{1}{\pi\tau_L}$$

For gas laser: collision broadening - pressure broadening

Pressure $\uparrow \rightarrow$ collision between atom (molecule) $\uparrow \rightarrow$ collision broadening $\uparrow$

$$\Delta\nu_L = \alpha p$$

α proportionality coefficient

CO₂: $\alpha=49$ kHz/Pa

p pressure

He³:Ne²⁰(7:1): $\alpha=750$ kHz/Pa

3) Natural Broadening + Collision Broadening: still Lorentz function

$$\Delta\nu_H = \frac{1}{2\pi} \left(\frac{1}{\tau_s} + \frac{2}{\tau_L} \right) = \Delta\nu_N + \Delta\nu_L$$

4) Phonon: Lattice Vibration Broadening

- Due to the thermal lattice vibration, the excited ions fixed inside the lattice that will change with time, which results in position change within a certain range, and therefore induces the line broadening.
- The thermal lattice vibration has the same effect to all luminous ions, and it is classified as homogenous broadening. Lattice vibration broadening is the main broadening mechanism for solid-state gain materials.

Homogenous Broadening—The physical mechanism which causes the line broadening is equivalent for all the atoms; Every light emitting atoms follow the line-shape of

Lorentz function

$$\tilde{g}_H(\nu, \nu_0) = \frac{\Delta\nu_H/2\pi}{(\nu - \nu_0)^2 + (\Delta\nu_H/2)^2}$$

1) Doppler Broadening

Light emitting from particles with thermal motion has line broadening due to Doppler frequency shift

light source

ν_0

ν

Receiver

$\nu_z < 0$ $\nu_z > 0$

$\nu = \nu_0 \sqrt{\frac{1 + \nu_z/c}{1 - \nu_z/c}}$

$\nu_z/c \ll 1$

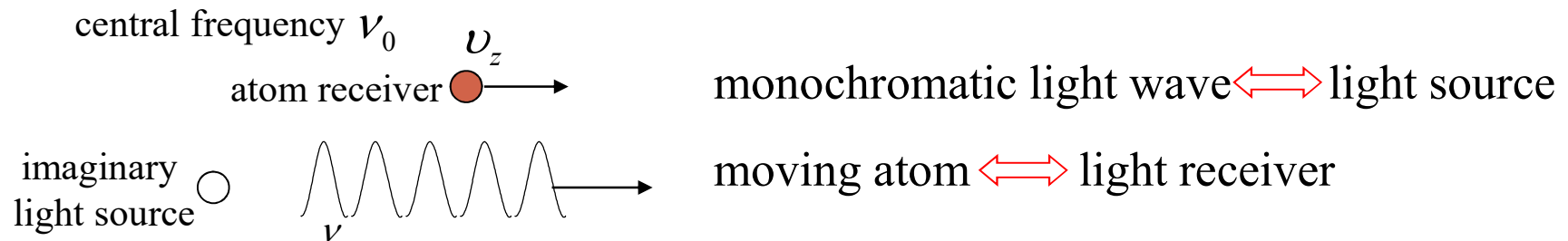
1st order approximation

$\nu = \nu_0 \left(1 + \frac{\nu_z}{c}\right)$

Atom moves toward receiver
(along the light propagation direction)

Atom leaves the receiver
(reverse direction of propagation)

Stimulated emission Doppler effect: one-dimensional Doppler effect



if $\nu_z = 0$ the frequency that atom feel is ν

biggest resonant interaction when $\nu = \nu_0$

when atom moving along z $\nu' = \nu(1 - \nu_z/c)$

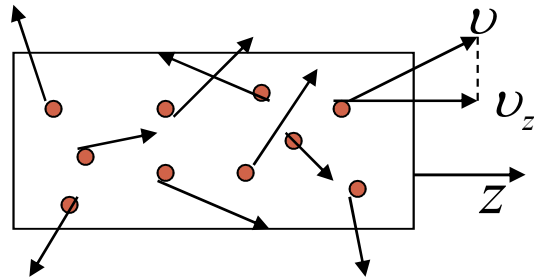
biggest resonant interaction when $\nu' = \nu_0$

$$\nu = \nu_0 / \left(1 - \frac{\nu_z}{c}\right) = \nu_0 \left(1 + \frac{\nu_z}{c}\right) / \left[1 - \left(\frac{\nu_z}{c}\right)^2\right] \approx \nu_0 \left(1 + \frac{\nu_z}{c}\right) = \nu'_0 \quad \left(\frac{\nu_z}{c} \ll 1\right)$$

Atoms with speed of ν_z interacts with wave propagates in the z interaction, the center frequency of atom ν'_0 is

$$\nu'_0 = \nu_0 \left(1 + \frac{\nu_z}{c}\right) \quad \begin{array}{ll} \nu_z > 0 & \text{The same direction of atom movement and light propagation} \\ \nu_z < 0 & \text{The reverse direction of atom movement and light propagation} \end{array}$$

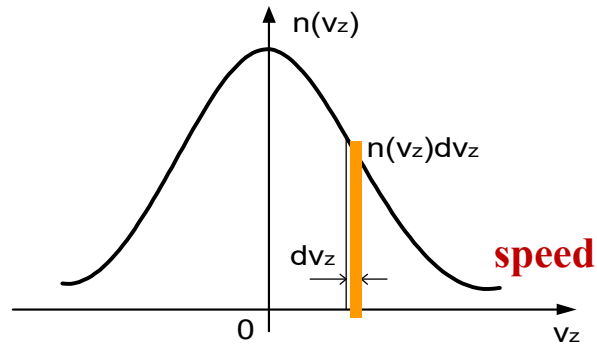
The Line Function of Doppler Broadening



1) Atom along z direction: Maxwell distribution

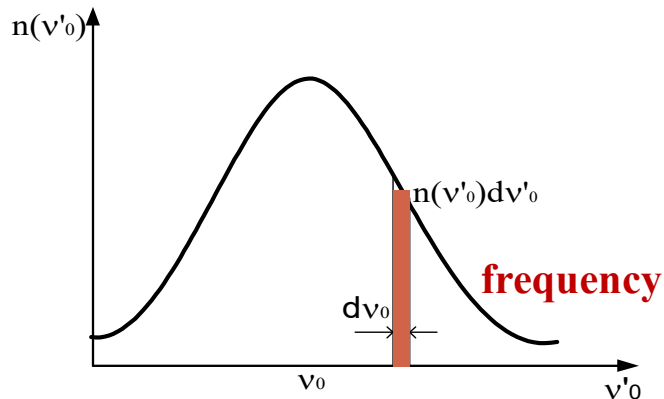
$$n(v_z)dv_z = n \left(\frac{m}{2\pi k_b T} \right)^{1/2} e^{-mv_z^2/2k_b T} dv_z$$

mass of atom



$$\text{for } E_2 \quad n_2(v_z)dv_z = n_2 \left(\frac{m}{2\pi k_b T} \right)^{1/2} e^{-mv_z^2/2k_b T} dv_z$$

$$\text{for } E_1 \quad n_1(v_z)dv_z = n_1 \left(\frac{m}{2\pi k_b T} \right)^{1/2} e^{-mv_z^2/2k_b T} dv_z$$



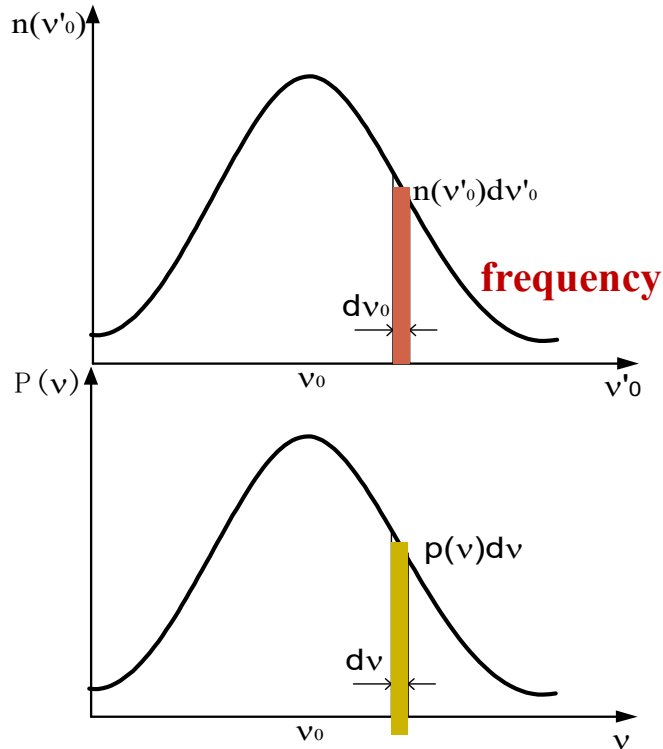
2) Atom around the central frequency

$$\because v'_0 = v_0 \left(1 + \frac{v_z}{c} \right) \quad dv_z = \frac{c}{v_0} dv'_0$$

$$\tilde{g}_D(v'_0, v_0) = \frac{n_2(v'_0)}{n_2} = \frac{n_1(v'_0)}{n_1}$$

$$n_2(v'_0)dv'_0 = n_2 \frac{c}{v_0} \left(\frac{m}{2\pi k_b T} \right)^{1/2} e^{-\frac{mc^2(v'_0 - v_0)^2}{2k_b T v_0^2}} dv'_0$$

Atom around the central frequency



$$\tilde{g}_D(\nu', \nu_0) = \frac{c}{\nu_0} \left(\frac{m}{2\pi k_b T} \right)^{1/2} e^{-\frac{mc^2}{2k_b T \nu_0^2} (\nu' - \nu_0)^2}$$

Doppler line broadening function

$$\tilde{g}(\nu, \nu_0) = \frac{P(\nu)}{P} \quad n(\nu_z) \rightarrow n(\nu'_0) \rightarrow n(\nu)$$

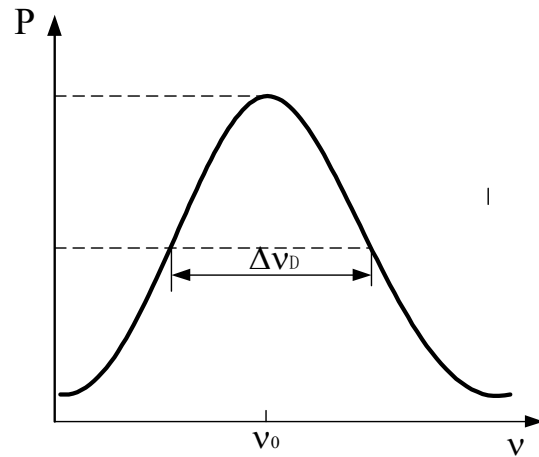
if does not consider $\Delta\nu_N, \Delta\nu_L$ the SP frequency $\nu = \nu'_0$

$$P(\nu) d\nu = h\nu_0 n_2(\nu) A_{21} d\nu = h\nu_0 A_{21} n_2 \tilde{g}_D(\nu, \nu_0) d\nu$$

$$P = n_2 A_{21} h\nu_0$$

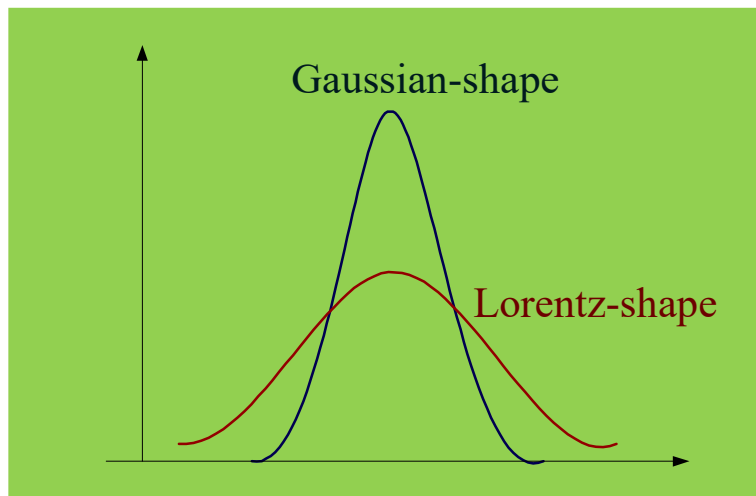
Gaussian function

$$\tilde{g}_D(\nu, \nu_0) = \frac{c}{\nu_0} \left(\frac{m}{2\pi k_b T} \right)^{1/2} e^{-\frac{mc^2}{2k_b T \nu_0^2} (\nu - \nu_0)^2}$$



Doppler broadening
line-width

均匀是洛伦兹
非均匀是高斯



$$\tilde{g}_D(\nu, \nu_0) = \frac{c}{\nu_0} \left(\frac{m}{2\pi k_b T} \right)^{1/2} e^{-\frac{mc^2}{2k_b T \nu_0^2} (\nu - \nu_0)^2}$$

$$\text{when } \nu = \nu_0 \quad \tilde{g}_D(\nu_0, \nu_0) = \frac{c}{\nu_0} \left(\frac{m}{2\pi k_b T} \right)^{1/2}$$

$$\Delta \nu_D = 2\nu_0 \left(\frac{2k_b T}{mc^2} \ln 2 \right)^{1/2} = 7.16 \times 10^{-7} \nu_0 \left(\frac{T}{M} \right)^{1/2}$$

atomic
weight

$$\tilde{g}_D(\nu, \nu_0) = \frac{2}{\Delta \nu_D} \left(\frac{\ln 2}{\pi} \right)^{1/2} e^{-\frac{4 \ln 2 (\nu - \nu_0)^2}{\Delta \nu_D^2}}$$

$$\tilde{g}_D(\nu_0, \nu_0) = \frac{2}{\Delta \nu_D} \left(\frac{\ln 2}{\pi} \right)^{1/2} = \frac{0.939}{\Delta \nu_D}$$

$$\tilde{g}_L(\nu_0, \nu_0) = \frac{2}{\pi \Delta \nu_N} = \frac{0.637}{\Delta \nu_N}$$

Doppler broadening in gas is belong to inhomogeneous broadening. The central frequency is different for atoms with different velocities, each one contributes to the spectral line and leads to broadening.

2) Local Field Inhomogeneous Broadening in Solid-State Materials

Crystal defects (dislocation, vacancy) caused the disorder of crystal field. The poorer the crystal quality, the wider the spectral line-width. It can not be described by any function, and can only be measured.

For glass materials doped with active ions, the activated ion located at inequivalent coordination field will lead to inhomogeneous broadening, such as neodymium glass.

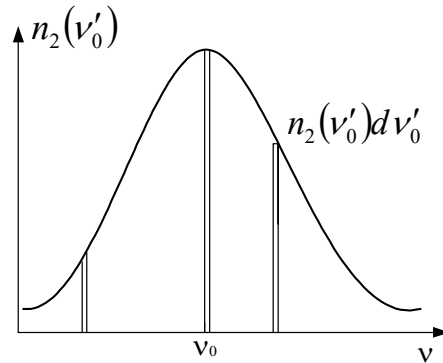
Differences between homogeneous and inhomogeneous broadening:

- Homogeneous broadening: emission from any light emitting particles (atoms, ions, molecules) contribute to every frequency in the spectrum
- Inhomogeneous broadening: emission from light emitting particles only contribute to certain frequency. In the inhomogeneous broadening, different light emitting particles contribute to different frequency inside $\tilde{g}(\nu)$

III. Comprehensive Broadening

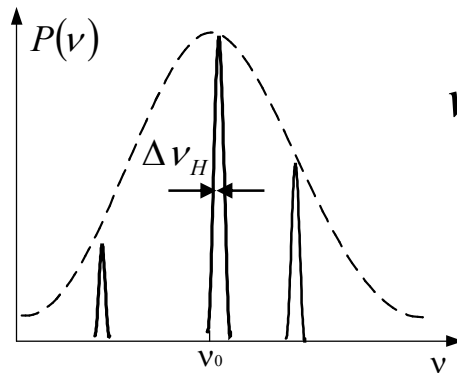
homogenous + inhomogeneous

The comprehensive line function of gas



1) $\nu'_0 \rightarrow \nu'_0 + d\nu'_0$ the SP from energy level E_2

$$\frac{n_2 \tilde{g}_D(\nu'_0, \nu_0) d\nu'_0 h\nu A_{21} \tilde{g}_H(\nu, \nu'_0) d\nu}{\nu'_0 \rightarrow \nu'_0 + d\nu'_0 \quad E_2 \text{ level } n_2(\nu'_0) \quad \text{SP possibility } \nu \rightarrow \nu + d\nu}$$



2) the SP from all the particle $\nu \rightarrow \nu + d\nu$

$$\nu \approx \nu_0 \quad P(\nu) d\nu = d\nu \int_{-\infty}^{\infty} n_2 \tilde{g}_D(\nu'_0, \nu_0) d\nu'_0 h\nu A_{21} \tilde{g}_H(\nu, \nu'_0)$$



$$P(\nu) = h\nu_0 n_2 A_{21} \int_{-\infty}^{\infty} \tilde{g}_D(\nu'_0, \nu_0) \tilde{g}_H(\nu, \nu'_0) d\nu'_0$$

$$P = \int_{-\infty}^{\infty} P(\nu) d\nu = h\nu_0 n_2 A_{21}$$

$$\tilde{g}(\nu, \nu_0) = \frac{P(\nu)}{P} = \int_{-\infty}^{\infty} \tilde{g}_D(\nu'_0, \nu_0) \tilde{g}_H(\nu, \nu'_0) d\nu'_0$$

Comprehensive
broadening line
function

$$\tilde{g}(\nu, \nu_0) = \int_{-\infty}^{\infty} \tilde{g}_D(\nu'_0, \nu_0) \tilde{g}_H(\nu, \nu'_0) d\nu'_0$$

The comprehensive broadening
line function is the convolution of $\tilde{g}_D(\nu'_0, \nu_0), \tilde{g}_H(\nu, \nu'_0)$

(1) $\Delta \nu_H \ll \Delta \nu_D$

$\tilde{g}_D(\nu'_0, \nu_0)$ can be regarded as δ function, only has non-zero value when $\nu'_0 \approx \nu$

$$\tilde{g}(\nu, \nu_0) = \tilde{g}_D(\nu, \nu_0) \int_{-\infty}^{\infty} \tilde{g}_H(\nu, \nu'_0) d\nu'_0 = \tilde{g}_D(\nu, \nu_0)$$

When $\Delta \nu_H \ll \Delta \nu_D$, the comprehensive broadening is mainly due to
Doppler inhomogeneous broadening

(2) When $\Delta \nu_D \ll \Delta \nu_H$ similarly, $\tilde{g}(\nu, \nu_0) = \tilde{g}_H(\nu, \nu_0)$

the comprehensive broadening is mainly due to
homogenous broadening

(1) He-Ne: 632.8 nm

$$\Delta\nu_N \approx 10^7 \text{ Hz}$$

$$\Delta\nu_L \approx 100 \sim 300 \text{ MHz}$$

$$\Delta\nu_D \approx 1500 \text{ MHz}$$

For He-Ne laser, Doppler inhomogeneous broadening is the main mechanism

(2) CO₂: 10.6 μm

$$\Delta\nu_N \approx 10^3 \sim 10^4 \text{ Hz}$$

$$\Delta\nu_L = \alpha p \quad \alpha = 49 \text{ kHz/Pa}$$

$$\Delta\nu_D \approx 60 \text{ MHz}$$

P ~ 1333 Pa comprehensive; P >> 1333 Pa homogenous

(3) Broadening of solid-state materials

➤ Ruby: Low temperature - inhomogeneous broadening

Room temperature - homogeneous broadening $2.7 \times 10^5 \text{ MHz}$

➤ Nd: YAG crystals: homogeneous broadening due to lattice thermal vibration $1.95 \times 10^5 \text{ MHz}$

➤ Nd: Glass: inhomogeneous broadening $7 \times 10^6 \text{ MHz}$

